Will Epperson

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Google Scholar

Education

Ph.D. in Human Computer Interaction August 2020 - June 2025
Carnegie Mellon University
Advisors: Dominik Moritz, Adam Perer
Thesis: Interactive Data Profiling

M.S. in Human Computer Interaction August 2020 - May 2023
Carnegie Mellon University
Advisors: Dominik Moritz, Adam Perer

B.S. in Computer Science August 2016 — May 2020
Georgia Institute of Technology
GPA: 4.0, Summa Cum Laude, threads in Intelligence and Modeling/Simulation

Industry Experience

Microsoft AI, Redmond, WA August 2026 - Present
Member of Technical Staff
Working on reinforcement learning environments and agentic post-training.

Microsoft Research, Redmond, WA August 2025 - August 2026
Senior Research Engineer, AI Frontiers
Research engineer working on AI agents, multi-agent systems, AI evaluations, and environments.

Microsoft Research, Redmond, WA Summer 2024
Research Intern, AI Frontiers - HAX Group
Mentor: Gagan Bansal, Victor Dibia
Research intern working on developer tools for multi-agent AI systems.

Databricks, San Francisco, CA Summer 2022
Software Engineering Contractor
Mentor: Kanit Wongsuphasawat
Designed and delivered production feature for creating dashboards by specifying fields of interest in a dataset.

Microsoft Research, Redmond, WA Summer 2021
Research Intern, VIDA Group
Mentor: Steve Drucker, Rob DeLine
Research intern working on data science tools. Lead project around reuse and sharing in data science, published at ICSE 2022. Co-author on project around visualizing data frame differences published at CHI 2022.

Point72 Asset Management, New York, NY Summer 2019
Data Analytics Intern, Market Intelligence Group
Mentor: Trevor Rempel
Worked as Data Scientist in alternative data space to clean, model, and understand large datasets

Ultimate Software, Weston, FL Summer 2018
Software Development Intern, Innovation Strategies Team
Mentor: Joseph Cutrono
Designed and developed Slack app to integrate with the UltiPro HR management tool. App published to Slack app store.

The Home Depot, Atlanta, GA — Summer 2015

Software Development Intern

Developed web app for tracking candidate progress throughout hiring process for internal HR use.

Research Experience

Carnegie Mellon University, Pittsburgh, PA — August 2020 - June 2025

Graduate Researcher, Data Interaction Group (DIG)

Advisor: Dominik Moritz, Adam Perer

Member of the DIG research group, working on novel data visualizations, ML interpretation techniques, and interactive data systems.

Georgia Institute of Technology, Atlanta, GA — January 2019 - May 2020

Undergraduate Researcher, Polo Club of Data Science

Advisor: Duen Horng (Polo) Chau

Member of the Polo Club of Data Science working on novel data visualizations to find fairness issues in Machine Learning models

Georgia Institute of Technology, Atlanta, GA — January 2018 - May 2019

Undergraduate Researcher, Automated Algorithm Design

Advisor: Jason Zutty, Greg Rohling

Worked on EMADE algorithm design engine to implement sentiment analysis pipeline to analyze news articles to aid in predicting stock price movements using genetic algorithms. Led project to visualize the genetic algorithm evolution process.

Publications

CAVEAT: Towards Robust Computer-Use Agents in Incentive-Misaligned Environments

Yuxuan Li, Will Epperson, Wesley Deng, Zezhou Huang

A benchmark for testing whether computer-use agents preserve user objectives when marketplaces steer them toward other products.

Preprint. 2026.

Project PDF Code

SocialReasoning-Bench: Measuring Whether AI Agents Act in Users' Best Interests

Tyler Payne, Will Epperson, Safoora Yousefi, Zachary Huang, Gagan Bansal, Wenyue Hua, Maya Murad, Asli Celikyilmaz, Saleema Amershi

A benchmark for measuring AI agents' negotiation ability in calendar coordination and marketplace settings.

Microsoft Research Blog. 2026.

Project Blog Code

Whimsical Strategies Break AI Agents: Generating Out-of-Distribution Adversarial Strategies at Scale

Zachary Huang, Tyler Payne, Gagan Bansal, Will Epperson, Wenyue Hua, Adam Fourney, Amanda Swearngin, Maya Murad, Ece Kamar, Saleema Amershi

Automatically generating out-of-distribution adversarial strategies that reliably manipulate frontier agents in negotiations.

Microsoft Research Blog. 2026.

Project Blog

Red-Teaming a Network of Agents: Understanding What Breaks When AI Agents Interact at Scale

Gagan Bansal, Shujaat Mirza, Keegan Hines, Will Epperson, Zachary Huang, Whitney Maxwell, Pete Bryan, Tyler Payne, Adam Fourney, Amanda Swearngin, Wenyue Hua, Tori Westerhoff, Amanda Minnich, Maya Murad, Ece Kamar, Ram Shankar Siva Kumar, Saleema Amershi

We evaluated a live network of over 100 always on AI agents and found new failure modes that only emerge at the network level.

Microsoft Research Blog. 2026.

Project Blog

Magentic Marketplace: An Open-Source Environment for Studying Agentic Markets
Gagan Bansal, Wenyue Hua, Zezhou Huang, Adam Fourney, Amanda Swearngin, Will Epperson, Tyler Payne, Jake M. Hofman, Brendan Lucier, Chinmay Singh, Markus Mobius, Akshay Nambi, Archana Yadav, Kevin Gao, David M. Rothschild, Aleksandrs Slivkins, Daniel G. Goldstein, Hussein Mozannar, Nicole Immorlica, Maya Murad, Matthew Vogel, Subbarao Kambhampati, Eric Horvitz, Saleema Amershi
An open-source simulation environment for studying how LLM agents behave as buyers and sellers in two-sided markets.
Preprint. 2025.
Project PDF Code

Interactive Data Profiling
Will Epperson
My PhD thesis on interactive systems to help users automatically understand their data while programming.
Carnegie Mellon University (CMU). Pittsburgh, PA, 2025.
Project PDF

Texture: Structured Exploration of Text Datasets
Will Epperson, Arpit Mathur, Dominik Moritz, Adam Perer
Texture is a general purpose text exploration tool with interactions for using LLMs to derive data from text.
Open source. 2025.
Project PDF Code

Interactive Debugging and Steering of Multi-Agent AI Systems
Will Epperson, Gagan Bansal, Victor Dibia, Adam Fourney, Jack Gerrits, Erkang Zhu, Saleema Amershi
AGDebugger is an interactive debugging tool for multi-agent AI systems.
2025 CHI Conference on Human Factors in Computing Systems (CHI). Yokohama, Japan, 2025.
Project PDF Code

Over-Relying on Reliance: Towards Realistic Evaluations of AI-Based Clinical Decision Support
Venkatesh Sivaraman, Katelyn Morrison, Will Epperson, Adam Perer
We discuss how evaluations of human-AI decision support systems can move beyond reliance as the primary metric.
2025 CHI Workshop on HCI+Health (CHI). Yokohama, Japan, 2025.
Project PDF

Guided Statistical Workflows with Interactive Explanations and Assumption Checking
Yuqi Zhang, Adam Perer, Will Epperson
GuidedStats is a Jupyter extension that helps data scientists perform statistical analyses with guided workflows.
VIS 24: IEEE Conference on Data Visualization (VIS). St Pete Beach, Florida, 2024.
Project PDF Code

Dead or Alive: Continuous Data Profiling for Interactive Data Science
Will Epperson, Vaishnavi Gorantla, Dominik Moritz, Adam Perer
AutoProfiler is a Jupyter extension that helps data scientists understand their data and find issues during analysis through continuous data profiling.
VIS 23: IEEE Conference on Data Visualization (VIS). Melbourne, Australia, 2023.
Project PDF Code Best Paper Honorable Mention

A Declarative Specification for Authoring Metrics Dashboards
Will Epperson, Kanit Wongsuphasawat, Allison Whilden, Fan Du, Justin Talbot
Quick dashboarding presents a novel specification for dashboard authoring, comprised of sections of metrics combined with dimensions.
VDS at VIS 23: Visual Data Science Symposium (VDS). Melbourne, Australia, 2023.
Project PDF Best Paper

Leveraging Analysis History for Improved In Situ Visualization Recommendation
Will Epperson, Doris Jung-Lin Lee, Leijie Wang, Kunal Agarwal, Aditya Parameswaran, Dominik Moritz, Adam Perer
Solas is a visualization recommendation tool that uses the history of analysis for in situ recommendations in Jupyter.
EuroVis 22: Eurographics Conference on Visualization (EuroVis). Rome, Italy, 2022.
Project PDF Code

Strategies for Reuse and Sharing among Data Scientists in Software Teams
Will Epperson, April Yi Wang, Robert DeLine, Steven M. Drucker
Interviews and a survey with 149 data scientists at Microsoft revealed five distinct strategies for sharing and reusing analysis code along with factors that encourage or discourage reuse.
ICSE 22: ACM International Conference on Software Engineering (ICSE). Pittsburgh, PA, 2022.
Project PDF Recording Slides

Diff in the Loop: Supporting Data Comparison in Exploratory Data Analysis
April Yi Wang, Will Epperson, Robert DeLine, Steven M. Drucker
Diff in the Loop supports tracking, comparing, and visualizing differences in datasets during iterative data analysis.
SIGCHI 22: ACM Symposium on Computer Human Interaction (CHI). New Orleans, LA, 2022.
Project PDF

RECAST: Interactive Auditing of Automatic Toxicity Detection Models
Austin P. Wright, Omar Shaikh, Haekyu Park, Will Epperson, Muhammed Ahmed, Stephane Pinel, Diyi Yang, Duen Horng (Polo) Chau
Interactive Auditing of Automatic Toxicity Detection Models
24th ACM Conference on Computer-Supported Cooperative Work & Social Computing. 2021.
Project PDF

FairVis: Visual Analytics for Discovering Intersectional Bias in Machine Learning
Angel Cabrera, Will Epperson, Fred Hohman, Minsuk Kahng, Jamie Morgenstern, Duen Horng (Polo) Chau
Discovering intersectional ML Bias through interactive visualization.
IEEE Conference on Visual Analytics Science and Technology (VAST). Vancouver, Canada, 2019.
Project Demo PDF Blog Recording Code

Talks

Interactive Data Profiling for Python DataFrames with AutoProfiler and Texture
PyCON 2024 — May 2024

Dead or Alive: Continuous Data Profiling for Interactive Data Science
VIS 23: IEEE Visualization Conference — October 2023

A Declarative Specification for Authoring Metrics Dashboards
VDS at VIS 23: Visual Data Science Symposium — October 2023

Leveraging Analysis History for Improved In Situ Visualization Recommendation
EuroVis 22: Eurographics Conference on Visualization — June 2022

Strategies for Reuse and Sharing among Data Scientists in Software Teams
ICSE 22: ACM International Conference on Software Engineering — May 2022

FairVis
VIS 19: IEEE Visualization Conference — October 2019

Honors and Awards

PURA: President's Undergraduate Research Award — 2019
$1500 research grant to continue work on FairVis project

Stamps President's Scholarship — 2016
Full ride scholarship given to 40 incoming freshman at Georgia Tech

Mentees

During my PhD, I have had the pleasure of mentoring the following undergraduate and masters students on research projects.

Leijie Wang — Summer 2021 - Fall 2021
Visualization recommendation for python in notebooks using history

Asad Sheikh Fall 2021 - Spring 2022

Visualization recommendation for SQL using history

Vaishnavi Gorantla Spring 2022 - Spring 2023

Fact generation from data and presentation as text

Yuqi Zhang Spring 2023 - Spring 2024

Notebook extension for guided statistical analysis

Allie Feldman Summer 2023 - Spring 2024

Guided Data Analysis

Teaching

Graduate Teaching Assistant Spring 2023

Carnegie Mellon University, Pittsburgh, PA

Programmable User Interfaces, Instructor: Scott Hudson

Taught recitation, designed assignments for class about UI design and intro HTML, CSS, & Javascript.

Graduate Teaching Assistant Fall 2022

Carnegie Mellon University, Pittsburgh, PA

Interactive Data Science, Instructor: Adam Perer and John Stamper

Graded and office hours for class about using jupyter, visualization, steamlit and related tech for data science.

Undergraduate Teaching Assistant August 2017 - December 2018

Georgia Institute of Technology, Atlanta, GA

Intro to Database Systems (CS 4400), Instructor: Monica Sweat

Designed projects, held office hours and graded for relational databases class.

Service

Reviewer for CHI 2025, VIS 2024, VIS 2023, CHI 2023, VIS 2022, CHI 2022, CSCW 2021, VIS 2021.

Leadership & Activities

Student Ambassador January 2019 - May 2020

Georgia Institute of Technology Alumni Association

Serve as official representative of the Institute at events/tours for alumni, prospective students, and special guests.

Executive Board Member -- Threads Co-chair January 2018 - May 2019

Stamps Scholars National Convention 2019

Executive board member of Stamps Scholars National Convention, a 3-day conference with over 700 student attendees. Responsible for 20-person committee that plans and coordinates the different content threads of the convention.

Skills

Programing Languages: Python (Advanced), Javascript/Typescript (Advanced), SQL (Intermediate)
Toolkits, Frameworks, Software: Svelte, React, Pytorch, Scikit-learn, Git, VegaLite, D3
Natural Languages: English (Native), Spanish (Advanced)